PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Is there a correlation between children's outdoor active mobility behaviour and neighbourhood safety? A systematic review of the evidence
AUTHORS	Zougheibe, Roula; Jepson, Beverly; Norman, Richard; Gudes, Ori; Dewan, Ashraf

VERSION 1 – REVIEW

REVIEWER	Vaalavuo, Maria National Institute for Health and Welfare
REVIEW RETURNED	16-Feb-2021

GENERAL COMMENTS	Review report Thank you for the opportunity to review this article. The objectives of the article were first to synthesize evidence on the associations between perceived and measured neighbourhood safety and children's active mobility, and second, to develop a context-specific methodological quality evaluation. The article is well-written, and considering the interest in neighbourhood context and health, childhood in the urban context, and growing inequalities and segregation, the article is well justified. I have a few comments that would improve the contribution of the article in my opinion. First, as neighbourhood context, and most probably also perceived and measured safety, is associated with various socioeconomic indicators, I would like to see a better discussion on the topic of inequalities related to children's living environment and access to safe outdoor activities. This is only briefly mentioned in the concluding section. This discussion and the contribution of the article could be connected to the various findings on socioeconomic inequalities in childhood physical activity levels and obesity as well as neighbourhood context. Connecting the discussion and review better to socioeconomic inequalities would be a further merit of the article. It would be a good addition to evaluate to what extent such important dimension has been considered in the reviewed articles. Or whether this could be highlighted as a topic that needs to be integrated in future studies. Although I am not an expert on the methodology of systematic reviews, I think the review has been conducted in an appropriate manner and all the stages have been well documented in the article. However, I was not really convinced about the quality evaluation. It is partly so context-specific that I am left wondering about the general usefulness of the criteria. Perhaps the authors could better motivate this part of the article and explain the purpose of the endeavor. In the introduction I got the impression that the aim was to develop a methodological tool to evaluate

	quality in other research too, but I was not so sure about this anymore when the criteria were discussed. In the synthesis of the evidence it would help the reader if the direction of the association was mentioned. What does 'strong association' with ethnic background mean? Negative or positive? And what does "ethnic" mean in this context when there are studies from different countries (e.g. Malaysia and Iran, what is ethnic there or even in the US or the UK, this is not clear). Ethnic minorities could work perhaps better? Also it is not clear what a 'negative association' with car ownership mean, you could start the paragraph by mentioning that you are looking at high level of activity, in other words, here car ownership is associated with a lower activity if I follow correctly? But I am uncertain of this interpretation here. Rather than just repeating "positive or negative association", it would be good to sometimes write "higher x is associated with lower y". The finding that perceived indicators yield different results from objective measures lead me to believe that this is not so much a question of neighbourhood context but parents' perceptions and characteristics in general. What do you think? In general, the review does not highlight any evidence that would actually be based on methodologically solid "neighbourhood effects" studies that would be able to make a 'causal link' between safety and activity. I think this should be acknowledged in the introduction and in the title of the article (replacing the word "impact" with association or correlation, for example). It should be acknowledged that studies on neighbourhood effects are full of methodological challenges, and especially with a cross-sectional study it is impossible to identify causality. Is there research on variation in childhood activity by neighbourhood types? I think this would be the starting point of this article to mention some of such findings. I would then link it to possibilities to play outdoor (green spaces, proximity of nature, etc.) and the obstacles for doing so (poor safety). I think the beginning of the article misses this narrative a bit. Some of the results in Table 1 seem counterintuitive: first study indicates that parental concerns over road safety are associated with higher walking levels. Or why is car ownership associated with walking (obviously I had understood this wrong above, and the direction is indeed the opposite, perhaps slightly counterintuitive)? Perhaps such results should be explained. I would add in the table whether the study was a multi-level study so that there were several individuals from the same neighbourhood answering the survey etc. What other factors measured at neighbourhood-level could explain the results, is safety just associated with lower SES of the neighbourhood? To what extent we have parents living in the same neighbourhood with opposite perceptions and differences in children's activity. I think the synthesis section should consider methodological issues better, to what extent they tell us anything about neighbourhoods? (The discussion and concluding sections are quite long and with some repetition, I would rather see a longer discussion on the reviewed articles and the synthesis.) Smaller comments: Abstract: what is active travel type? Could be clarified here if used. It is not clear what socio-ecological framework here means or socio-ecological level. Activity measured solely by BMI seems like a weak indicator, the inclusion of studies using only BMI should be justified and

	discussed or better yet removed as BMI is a result of so much more than activity. Why were the majority of the studies included of poor quality? Any suggestions on how to improve quality of future research? There are quite a lot of figures and tables that can be quite exhausting. Are all really necessary? Could some of the information regarding the included studies be summarized in a table rather than dozens of figures?

REVIEWER	Hassani, Kasra Provincial Health Services Authority, Child Health BC
REVIEW RETURNED	05-Apr-2021

GENERAL COMMENTS	The manuscript by Zougheibe et al. systematically reviews the relationship between children's active outdoor mobility and neighborhood safety. This is an important topic and worthy of review as it is important to gather evidence and make recommendations to improve children's outdoor mobility and activity. The manuscript has many strengths, namely the authors have been very thorough and diligent in their review, data extraction, coding, and synthesis. They provide useful implications and recommendations for improvement. However, the manuscript has a number of shortcomings that need to be addressed before I could recommend it for publications. My major concerns are with the methodology, eligibility criteria, and quality appraisal, which I have discussed below. The manuscript could also strongly benefit from English language editing as it would improve its understandability and fix its grammatical errors. Methodology: 1. Eligibility criteria: The study's time period is identified in the abstract but not in the methods section. 2. Please identify people who reviewed the abstracts by their initials. 3. Supplemental File 2 should begin with the search terms used. 4. Search strategy in other databases is missing. Only Scopus has been included. In a systematic review, the complete search strategy should be reproducible by the reader. Eligibility: It is not clear what type of publications have been included in the study. It appears that the others have only included publications where some sort of data has been directly collected, but more detail needs to be provided. Were other reviews (systematic, scoping, narrative, etc.) included or excluded? If so, how were they appraised? How was evidence from them incorporated? What about opinion pieces, editorials, and other similar non-original research publications? Also, did the authors make a distinction between experimental versus non-experimental (observational) studies? Quality appraisal: While there is a myriad of quality appraisal tools available in the literature, suitable for different study designs and review processes, the authors have created their own quality appraisal tool. I understand that the quality appraisal tool has been built specifically for the purpose of this review and looks at factors relevant to the study. Therefore, the criteria and point scores have

	some face validity but the instrument needs to go through its own review processes to be validated, before it can be utilized as a suitable and appropriate quality appraisal tool. For example, it appears to me that the quality appraisal tool is not adequately assessing possibility of bias. It also is not making any distinction between different types of evidence (experimental study design versus observational and also different kinds of each) as the strength of evidence derived from each of these designs strongly varies. Therefore, it is not clear if all pertinent quality characteristics of studies have been assessed in this tool. In addition, the scoring cut-offs for determining low/medium/high quality studies is rather arbitrary and not based on any evidence. I strongly recommend the authors to use a previously utilized and published quality appraisal tool in comparison with the current tool address this current tool's shortcomings, and provide a rationale of why it would be preferred. Synthesis: The authors need to provide a rationale for how they have chosen to synthesize the results, for example why a minimum of five studies were used for establishing a consensus? Discussion: The text has many grammatical errors and includes many convoluted sentences which fail to bring the point across. Although there are grammar issues throughout the manuscript, this shows most strongly in the discussion section. I strongly suggest the authors seek help from an editor to fix the grammar and make the text more readily accessible to the reader. Implications and conclusion: I find both sections a bit long and repetitive. I think they need to be shortened and be a bit more concise, especially the conclusion should not be longer than a single paragraph, highlighting the manuscript's key finding and key message. Figures: Overall: the # symbol is not appropriate for use in a manuscript. I suggest using the word Number, especially that there is no shortage of space here. Figure 3: I suggest for the first category (access to car), an empty column is created as placeholder for non-significant studies (I believe there weren't any?) Figure 4 is trying to show too much data in one place. Is it necessary to display all of these different dimensions? What is this Figure trying to show? I suggest it is reworked and simplified. Figure 5: It makes no mathematical and logical sense to have a y-axis that count number of studies and then plot -1 on it. Please rework this graph. This graph is very confusing and its purpose is unclear. Minor grammar comments: I started listing the grammar issues, but then there was so many that I had to stop. 1. Page 4, Line 12: Stranger danger, quotation missing. 2. Page 4, Line 16: Parents or children's perception of risk 3. Page 4, Line 20: Limited works were found 4. Page 4, Line 22: such as school travel?

	5. Page 4, Line 50: The word rigorous is not needed. A systematic search in itself implies rigor. 6. Page 5, Line 54: Do you mean rigorous? 7. Page 6, Line 44: Use term publication instead of manuscript. 8. Page 8, Line 42: I believe SES needs to be in parentheses. 9. Page 8: Line 37: Grammar. Please rewrite sentence. 10. Page 9: First sentence is unclear.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1

Thank you for the opportunity to review this article.
The objectives of the article were first to synthesise evidence on the associations between perceived and measured neighbourhood safety and children's active mobility, and second, to develop a context-specific methodological quality evaluation.
The article is well-written, and considering the interest in neighbourhood context and health, childhood in the urban context, and growing inequalities and segregation, the article is well justified.
I have a few comments that would improve the contribution of the article in my opinion.

Response: The authors would like to thank you for all the constructive feedback and the valuable comments that significantly improved various aspects of the paper.

	Comments by reviewers	**Action taken by authors**
Reviewer 1		
General remark Discussion	First, as neighbourhood context, and most probably also perceived and measured safety, is associated with various socioeconomic indicators, I would like to see a better discussion on the topic of inequalities related to children's living environment and access to safe outdoor activities. This is only briefly mentioned in the concluding section. This Discussion and the contribution of the article could be connected to the various findings on socioeconomic inequalities in childhood physical activity levels and obesity as well as neighbourhood context. Connecting the Discussion and review better to socioeconomic inequalities would be a further merit of the article. It would be a good addition to evaluate to what extent such important dimension has been considered in the reviewed articles. Or whether this could be highlighted as a topic that needs to be integrated in future studies.	Thank you for suggesting more focus on inequality related to children's active behaviour, which is evident when synthesising studies' findings. We have amended the revised manuscript with more emphasis on inequality. In the revised manuscript, we addressed inequality in children active behaviour at various aspects of individual, family and neighbourhood. In the Discussion (we addressed three topics around findings where inequality is the first topic). In Implication of Findings (we concluded intervention should address inequality but also in advancing research to address research gap) and in Conclusion sections. In Discussion, inspired by your argument, we formed a subheading of "**Inequality in opportunity for children's active behaviour"** (Please see page 8, lines 8 to 45) We suggested in **Implications of Findings**: interventions that address such inequality (Please see page 9 lines 39-47) In "**Advancing Research**", we suggested better measures in future studies to allow synthesis of evidence besides addressing

		knowledge gap in future research are crucial. (Please see page 10, lines 16 -26)
Quality	Although I am not an expert on the methodology of systematic reviews, I think the review has been conducted in an appropriate manner and all the stages have been well documented in the article. However, I was not really convinced about the quality evaluation. It is partly so context-specific that I am left wondering about the general usefulness of the criteria. Perhaps the authors could better motivate this part of the article and explain the purpose of the endeavor. In the introduction I got the impression that the aim was to develop a methodological tool to evaluate quality in other research too, but I was not so sure about this anymore when the criteria were discussed.	The authors have updated the section that addresses studies quality in the revised manuscript to eliminate any uncertainty and ensure clarity. In the Introduction section, we highlighted that "With the absence of suitable checklists assessing quality of observational studies, increased research calls for a valid approach to increase confidence in the synthesis of studies outcome were found. Majority of existing criteria evaluates studies reporting. Accounting for additional items in the checklist on measurement methods of observational studies may better inform evidence-based policy decisions. (Please see page 3, lines 30-33) We also explained that the quality tool was justified by previously published reviews addressing some shapes of children active behaviour. We used the tool from Schoeppe et al. (2013) and Marzi et al. (2018). We followed Scheoppe et al. (2013) method and expanded a few criteria that addressed the actual practices of measurement characteristics. We explained that the additional criteria resulted from a comprehensive review published by Zougheibe et al. 2021 on measurement methods for studies that examined active behaviour and neighbourhood safety. We reflect this in the revised "Quality assessment" (Please see page 5, lines 1-21).
Evidence Synthesise	In the synthesis of the evidence it would help the reader if the direction of the association was mentioned.	To avoid confusion and ensure clarity in the synthesis, we followed your suggestion, "it would be good to sometimes write "higher x is associated with lower y" Please see the revised manuscript Synthesis of the evidence (Please see page 7, lines13-50, page 8, lines 1-3)
	What does 'strong association' with ethnic background mean? Negative or positive?	In the revised manuscript, we clarified that there is inconsistency in the subgrouping of people ethnic backgrounds across studies accounting for race/ethnicity. This variability prevented drawing clear conclusions of direction despite indications of correlates. The change of active behaviour by ethnic background does exist yet varies in the direction by geographical location. We elaborated further in the synthesis of evidence and in Discussion.

		(Pease see Page 7 lines 25-30) (Pease see Page 8 lines 22-33)
	And what does “ethnic” mean in this context when there are studies from different countries (e.g. Malaysia and Iran, what is ethnic there or even in the US or the UK, this is not clear). Ethnic minorities could work perhaps better?	In the revised manuscript, we elaborated on how variation in geographic location correlates with variability in the ethnic/race grouping. And minority as subgrouping is used in two studies with the disagreed direction of correlates. (Pease see Page 7 lines 25-30) Corresponding with that, we listed in Table 1 supplemental file 2, each study accounted for ethnicity, the subgrouping of ethnic background under “Measures; Study accounted cofounders and other examined variables”.
	Also it is not clear what a ‘negative association’ with car ownership mean, you could start the paragraph by mentioning that you are looking at high level of activity, in other words, here car ownership is associated with a lower activity if I follow correctly? But I am uncertain of this interpretation here. Rather than just repeating “positive or negative association”, it would be good to sometimes write “higher x is associated with lower y”.	In the revised manuscript under “Synthesis of evidence” and across Individual, Family and Neighbourhood levels, we replaced negative or positive words with the output of correlates (e.g. lower/decreased or higher/increased active behaviour) (Please see page 7, lines 14-50 and page 8 lines 1-4)
	The finding that perceived indicators yield different results from objective measures lead me to believe that this is not so much a question of neighbourhood context but parents’ perceptions and characteristics in general. What do you think?	This is an area that requires further work. We found inconsistency in the method of measures among studies that used objective measure. Additionally, scarcity in studies examining variation between neighbourhood actual crime and perception of safety prevent giving a definite answer to such an important topic. We found a study by Kneedaw_Price et al. (2015) examined the variation of measured and perceived safety. Findings showed that lower physical activity was associated with neighbourhood crime but not with parents’ perception of neighbourhood safety. But with the scarcity of research addressing this correlate, further research may profoundly reveal variability between real and perceived crime. We addressed the issue in “Implications” under “Advancing Research.”

		(Please see page 10, lines 24-26)
	In general, the review does not highlight any evidence that would actually be based on methodologically solid "neighbourhood effects" studies that would be able to make a 'causal link' between safety and activity. I think this should be acknowledged in the introduction and in the title of the article (replacing the word "impact" with association or correlation, for example).	In the revised manuscript, we have modified the title as suggested. The title is now (**Is there a correlation between children's outdoor active mobility behaviour and neighbourhood safety? A systematic review of the evidence)** (Please see page 1, lines 1-2)
	It should be acknowledged that studies on neighbourhood effects are full of methodological challenges, and especially with a cross-sectional study it is impossible to identify causality.	Thank you for highlighting these critical challenges in the current research. In agreement with your argument, Zougheibe et al. has recently published a review paper on "Children's outdoor active mobility behaviour and neighbourhood safety: a systematic review in measurement methods and future research directions" in January 2021. The review builds upon synthesising individual studies measurements methods, illustrated existing challenges to highlighted future research directions. We have briefly added information on these challenges under the "Methodological challenges" of the "Implication of Finding" section. We also referred to the paper for further reading. (Please see page 9, lines 11-33)
Introduction	Is there research on variation in childhood activity by neighbourhood types? I think this would be the starting point of this article to mention some of such findings. I would then link it to possibilities to play outdoor (green spaces, proximity of nature, etc.) and the obstacles for doing so (poor safety). I think the beginning of the article misses this narrative a bit.	We conducted further research and reflected upon neighbourhood types and safety. We revised the introduction to reflect upon your suggestion (Please see page 3, lines 6-15). We also addressed findings on neighbourhood types in Results, Discussion, Implication on Findings and Conclusion

Table 1	Some of the results in Table 1 seem counterintuitive: first study indicates that parental concerns over road safety are associated with higher walking levels.	This has been revised in Table 1 of supplementary file 2. The unexpected correlate was clarified in the table and explained. The authors (Alton et al. 2017) suggested that parents might often warn children who are high walker about road danger, and therefore, children perceive roads as dangerous elements.
	Or why is car ownership associated with walking (obviously, I had understood this wrong above, and the direction is indeed the opposite, perhaps slightly counterintuitive)? Perhaps such results should be explained.	We revised Table 2 in supplemental file 2 to eliminate ambiguity for all studies. In general, each paper that examined cars ownership influence on children active travel has found a negative effect (i.e., lowered active behaviour). We updated studies that examined car ownership to eliminate ambiguity.
	I would add in the table whether the study was a multi-level study so that there were several individuals from the same neighbourhood answering the survey etc.	We revised table 1 in supplementary file 2. We added to Measures "**Safety (Perceived by)/** **Measured/** **Level of study" (here we listed all other variables examined in each study)** (Please see Table 1 in supplementary file 2)
	What other factors measured at neighbourhood level could explain the results, is safety just associated with lower SES of the neighbourhood?	We listed in table 1 in supplementary file 2 under Measures, a "Study accounted cofounders and other examined variables" each study addressed variables.
	To what extent we have parents living in the same neighbourhood with opposite perceptions and differences in children's activity.	This is a legitimate question that yet research has to find an answer. In the synthesised studies, (Timperio et al., 2004) found a variation of parents' perception of safety to their children. Scarcity in studies suggests addressing this topic in future research may offer a deeper understanding on how safety is perceived among parents comparing to their children. We revised the manuscript to highlight this research gap under methodological challenges (please see page 9, lines 30-31 Under "Implications of Findings -Advancing Research." (Please see page 10, lines 23-25)
	I think the synthesis section should consider **methodological issues better**, to what extent they tell us anything about neighbourhoods?	We addressed briefly the issue of accounting of child's neighbourhood. Under Discussion "Methodological Challenges" and in "Implications of Findings: Advancing

		research" current research measurements challenges. (Please see page 9, lines 23-29) (Please see page 10, line 24-26)
	(The discussion and concluding sections are quite long and with some repetition, I would rather see a longer discussion on the reviewed articles and the synthesis.)	We revised the Implications of Findings and reduced the Conclusion Section to one paragraph. (Please see Implication of Findings pages 9-10 and Conclusion, page 11, lines 8-26) The Discussion went under major revision to eliminate repetition (Please see Discussion from pages 8 line 6 to page 9 line 33).
Smaller comments:	Abstract: what is active travel type? Could be clarified here if used.	We clarified this in the revised manuscript (Please see page 2, line 15)
	It is not clear what socio-ecological framework (corrected line 9-12 page 2) here means or socio-ecological level.	We removed the phrase from the abstract, but in the revised manuscript (please page 4, line 40-44), we clarified the socio-ecological framework concept
	Activity measured solely by BMI seems like a weak indicator, the inclusion of studies using only BMI should be justified and discussed or better yet removed as BMI is a result of so much more than activity.	We agree with your point of view; however, we would like to note that the two studies that used BMI were (1) a longitudinal study, and (2) also used self-reported physical activity. We, thus, chose to have them remain in the review,
	Why were the majority of the studies included of poor quality? Any suggestions on how to improve quality of future research?	The author Zougheibe et al. published a paper in the Journal of Health Geographics titled "Children's outdoor active mobility behaviour and neighbourhood safety: a systematic review in measurement methods and future research directions". We referred to the paper's findings in the revised manuscript but briefly summarised current challenges See Methodological challenges (Please see page 9, lines 16-26). And in the measure and suggested future guidelines that could increase reliability in the evidence, "Advancing research". (Please see page 10, lines 15-26)

	There are quite a lot of figures and tables that can be quite exhausting. Are all really necessary? Could some of the information regarding the included studies be summarised in a table rather than dozens of figures?	Figures 5 and 6 were removed from the revised manuscript. In supplementary file 2, we have now four Tables and four diagrams. Although creating a table is visible to replace. However, for this specific analysis, we found graphs were a better visual tool to understand the studies strength and distribution across various active behaviour type of studies. Yet, we also reduced the number of diagrams in there.

Reviewer: 2

Dr. Kasra Hassani, Provincial Health Services Authority
Comments to the Author:
The manuscript by Zougheibe et al. systematically reviews the relationship between children's active outdoor mobility and neighborhood safety. This is an important topic and worthy of review as it is important to gather evidence and make recommendations to improve children's outdoor mobility and activity.
The manuscript has many strengths, namely the authors have been very thorough and diligent in their review, data extraction, coding, and synthesis. They provide useful implications and recommendations for improvement. However, the manuscript has a number of shortcomings that need to be addressed before I could recommend it for publications.
My major concerns are with the methodology, eligibility criteria, and quality appraisal, which I have discussed below. The manuscript could also strongly benefit from English language editing as it would improve its understandability and fix its grammatical errors.

Response: The authors would like to thank the reviewer for the critical comments that substantially improved the manuscript.

	Comments by reviewers		**Action taken by authors**
Reviewer 2			
Comment 1 Methodology	1.	Eligibility criteria: The study's time period is identified in the abstract but not in the methods section.	This has been included in the revised manuscript under search strategy (Please see page 4, line 4).
	2.	Please identify people who reviewed the abstracts by their initials.	This has been included in the revised manuscript. (Please see page 4, lines 22 – 26).
	3.	Supplemental File 2 should begin with the search terms used.	We moved search terms to the beginning f the file (Please, see Supplementary File 2, page 1).
	4.	Search strategy in other databases is missing. Only Scopus has been included. In a systematic review, the complete search strategy should be reproducible by the reader.	Other databases search strategies added to the revised Supplementary file2. (Please see supplementary file 2, pages 1-2).

Eligibility:	It is not clear what type of publications have been included in the study. It appears that the others have only included publications where some sort of data has been directly collected, but more detail needs to be provided.	We updated that in the revised manuscript. We only addressed peer-reviewed papers that data has been directly collected. We also clarified that intervention studies were excluded. (Please see Eligibility section page 4, lines 11-19).
	Were other reviews (systematic, scoping, narrative, etc.) included or excluded? If so, how were they appraised? How was evidence from them incorporated? What about opinion pieces, editorials, and other similar non-original research publications?	We clarified in the revised manuscript that review papers, opinion or other similar research publications were excluded. Please see Eligibility section page 4, lines 18-19).
	Also, did the authors make a distinction between experimental versus non-experimental (observational) studies?	Thanks for bringing this point up to recognise the type of papers included to avoid confusion. In the revised manuscript, we clarified that the studies included are observational. We mentioned that other kinds of studies, such as interventional, are excluded (Please see Eligibility section, page 4, lines 12-13).
Quality appraisal:	While there is a myriad of quality appraisal tools available in the literature, suitable for different study designs and review processes, the authors have created their own quality appraisal tool. I understand that the quality appraisal tool has been built specifically for the purpose of this review and looks at factors relevant to the study. Therefore, the criteria and point scores have some face validity but the instrument needs to go through its own review processes to be validated before it can be utilised as a suitable and appropriate quality appraisal tool. For example, it appears to me that the quality appraisal tool is not adequately assessing possibility of bias.	We updated the methodological quality assessment section to eliminate the ambiguity that maybe existed in the earlier version. We adopted a quality assessment tool justified by previously published systematic reviews. First, the criteria were used by an earlier review that synthesised evidence in correlating children's various types of active behaviour. The checklist covered criteria used by Scheoppe et al. (2013) and Mirza et al. (2018) that synthesised evidence of correlation to children independent mobility and Lubans et al. (2011) for children active travel. However, beyond criteria assessing studies reporting, there is an absence of a 'suitable checklist that assess quality characteristics of observational studies' as suggested by Schoeppe et al. We agree with the above finding confirmed in a systematic review of methodological measurement methods by Zougheibe et al. (2021). We, thus, expanded on six criteria that were an outcome of the methodological evaluation of measures (Zougheibe et al., 2021). The review addressed measurements methods in neighbourhood safety and correlated to children active behaviour. Lacking this checklist potentially affect studies reliability. Second, we assessed the robustness of each study, adopting Scheoeppe et al. (2013) and Mirza et al (2018), and adopted Scheoppe et al. for the cut-off for robust, moderate, and poor. (Please see page 5, line 1-21)

		Thirdly, we adopted Ikeda et al. 2018 and Rothman et al. 2018 to assign variables to well-defined categories. The authors also used a minimum of 5 studies agreeing in the direction of significance or non-significance to generate strength of evidence. (Please see evidence synthesis strength page 6 lines 1-8)
	It also is not making any distinction between different types of evidence (experimental study design versus observational and also different kinds of each) as the strength of evidence derived from each of these designs strongly varies.	We revised the manuscript to clarify that observational studies were included only in this work. (Please see page 4. line 13)
	Therefore, it is not clear if all pertinent quality characteristics of studies have been assessed in this tool.	The methodological quality assessment was revised to clarify each step of the methodological quality assessment. Starting from criteria checklist, scoring for identified criteria as well as calculating the percentage and assign strength. All mentioned are justified by previously published review papers (e.g., Schoeppe et al., (2011) and Mirza et al., (2018), Lubans et al. (2018) for assigning Robust, moderate, and poor. Please see the revised version (Please see page 5, lines 1-21).
	In addition, the scoring cut-offs for determining low/medium/high quality studies is **rather arbitrary and not based on any evidence.** I strongly recommend the authors to use a previously utilised and published quality appraisal tool in comparison with the current tool address this current tool's shortcomings, and provide a rationale of why it would be preferred.	
Synthesis:	The authors need to provide a rationale for how they have chosen to synthesise the results, for example why a minimum of five studies were used for establishing a consensus?	As for evidence strength, we clarified the revised manuscript that we adopted from Ikeda et al., 2018 and Rothman et al. (2018) reviews. The reviews carried vote counting and synthesise the evidence correlates when accumulated five to indicate a consensus sufficiently. (Please see page 6 lines 1-8)
Discussion:	The text has many grammatical errors and includes many convoluted sentences which fail to bring the point across. Although there are grammar issues throughout the manuscript, this shows most strongly in the discussion section. I strongly suggest the authors seek help from an editor to fix the grammar and make the text more readily accessible to the reader.	We apologise for the unintentional mistakes. They all have been corrected or eliminated from the text. The revised manuscript has gone through proof editing and proofreading. The primary author invited a BJ as native English speaker, to collaborate in the process of proofreading. Additionally, the author employed editing software for academic writing (Grammarly). The software is powerful in identifying faults in writing, such as a misspelled word, adding or removing commas, or grammatical and sentence structure. An editor then scanned the manuscript as a final check. Additionally, the mentioned phrase is now removed from the revised manuscript. Corrected

Implications and conclusion:	I find both sections a bit long and repetitive. I think they need to be shortened and be a bit more concise, especially the conclusion should not be longer than a single paragraph, highlighting the manuscript's key finding and key message.	We revised Implication of Findings and Conclusion sections to ensure that it is concise, and eliminate repetition. (Please see page 9-10 for implication) The "Conclusion" is also reduced now to one paragraph to bring the main key message of the review. (Please see pages 10-11 for Conclusion)
Figures:	Overall: the # symbol is not appropriate for use in a manuscript. I suggest using the word Number, especially that there is no shortage of space here.	We amended the suggested change in the revised manuscript figures and supplementary file diagram (Please see Figures 2,3, and 4 in the revised manuscript).
	Figure 3: I suggest for the first category (access to car), an empty column is created as placeholder for non-significant studies (I believe there weren't any?) Figure 4 is trying to show too much data in one place. Is it necessary to display all of these different dimensions? What is this Figure trying to show? I suggest it is reworked and simplified.	Thanks for your suggestion. We reflected this in the revised manuscript. We simplified Figure 4 and removed figures 5-6 from the revision. (Please see Figure 4)
	Figure 5: It makes no mathematical and logical sense to have a y-axis that count number of studies and then plot -1 on it. Please rework this graph. This graph is very confusing and its purpose is unclear.	
Minor grammar comments:	I started listing the grammar issues, but then there was so many that I had to stop. 1. Page 4, Line 12: Stranger danger, quotation missing. 2. Page 4, Line 16: Parents or children's perception of risk 3. Page 4, Line 20: Limited works were found 4. Page 4, Line 22: such as school travel? 5. Page 4, Line 50: The word rigorous is not needed. A systematic search in itself implies rigor. 6. Page 5, Line 54: Do you mean rigorous?	We apologise for the unintentional mistakes. They all have been corrected or eliminated from the text. The revised manuscript has gone through proof editing and proofreading. The primary author invited a new author BJ as native English speaker, to collaborate in the process of proofreading. Additionally, the authors employed editing software for academic writing (Grammarly). The software is powerful in identifying faults in writing, such as a misspelled word, adding or removing commas, or grammatical and sentence structure. An editor then scanned the manuscript as a final check. Additionally, the mentioned phrase is now removed from the revised manuscript. Corrected Corrected Corrected Corrected to active school travel The word rigorous is deleted. Corrected

	7. Page 6, Line 44: Use term publication instead of manuscript. 8. Page 8, Line 42: I believe SES needs to be in parentheses. 9. Page 8: Line 37: Grammar. Please rewrite sentence. 10. Page 9: First sentence is unclear.	Corrected Corrected

VERSION 2 – REVIEW

REVIEWER	Vaalavuo, Maria National Institute for Health and Welfare
REVIEW RETURNED	23-Jun-2021

GENERAL COMMENTS	I think all my comments have been taken into account well in the revised manuscript that has undergone a great deal of revision, thank you for considering my suggestions. The results from earlier studies are quite mixed so it's hard to sometimes follow the narrative. However, I think the manuscript has improved a lot and there are much less unclear sections.